Homework 8

MEGN544/ROBO554 Robot Mechanics: Kinematics, Dynamics, and Control

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Problem 1. Consider a three-link manipulator with the following DH-table.

Link	a	d	α	θ
1	0	L_1	$\pi/2$	$\star$
2	L_2	0	0	$\star$
3	L_3	0	0	$\star$

(a) Derive the forward kinematic equations symbolically for the position of the tool in the base frame as a function of θ_1, θ_2, and θ_3.

(b) Symbolically differentiate these kinematic equations to construct a symbolic representation of the position rows of the velocity Jacobian.

(c) Using the method outlined in class, calculate ${}^0\boldsymbol{z}_{k-1}$ and ${}^0\boldsymbol{d}_{k-1,N}$ for all $k = 1, \ldots, N$ symbolically and construct the position rows of the velocity Jacobian.

(d) For $L_1 = 1\,\text{m}$, $L_2 = 0.5\,\text{m}$, $L_3 = 0.3\,\text{m}$, $\theta_1 = 0\,\text{rad}$, $\theta_2 = \pi/4\,\text{rad}$, and $\theta_3 = \pi/4\,\text{rad}$ numerically evaluate your results from (b) and (c). *Hint: They should match!*